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EXPERIMENT 7

Aim:
Study and Implement of Security as a Service (SECaaS)

Theory:
1. What are different security issues in cloud computing?
Cloud computing stores data and applications on remote servers, which introduces several security risks.
Major Cloud Security Issues
- Data Breaches – Unauthorized access to sensitive information
- Account Hijacking – Stolen credentials and phishing attacks
- Insecure APIs – Weak interfaces used to access cloud services
- Misconfiguration – Incorrect permissions and open storage buckets
- Data Loss – Accidental deletion or disasters
- Insider Threats – Malicious actions by employees
- DDoS Attacks – Overloading servers to make services unavailable [1]

2. Explain Server and Data Security is cloud computing.
Server Security
Server security protects cloud virtual machines and infrastructure.
Methods used:
- Firewalls and security groups
- OS patching and updates
- Intrusion detection systems
- Network isolation using VPC

Data Security
Protects data stored in cloud services.
Methods used:
- Data encryption at rest and in transit
- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Backup and disaster recovery [2]

3. Explain various Threat detection, Data protection and Infrastructure protection services in AWS.
Threat Detection Services
• Amazon GuardDuty – Detects suspicious activities [3]
• AWS Shield – Protects from DDoS attacks [3]
• AWS WAF – Web application firewall [3]
Data Protection Services

• AWS KMS – Encryption key management [3]
• Amazon Macie – Finds sensitive data [3]
• AWS Secrets Manager – Secure credential storage [3]

Infrastructure Protection Services

• Amazon VPC – Network isolation [3]
• AWS IAM – Access control [3]
• AWS CloudTrail – Logs user activities [3]

4. What is Amazon VPC and its features?

Amazon Virtual Private Cloud (VPC) is a secure virtual network inside AWS used to launch cloud resources securely [2].

VPC Features

• Private and public subnets [2]
• Internet Gateway for public access [2]
• Route tables for traffic control [2]
• Security Groups (firewall) [2]
• Network ACLs [2]
• Elastic IP addresses [2]

Activity:

1. Build your VPC and Launch a Web Server

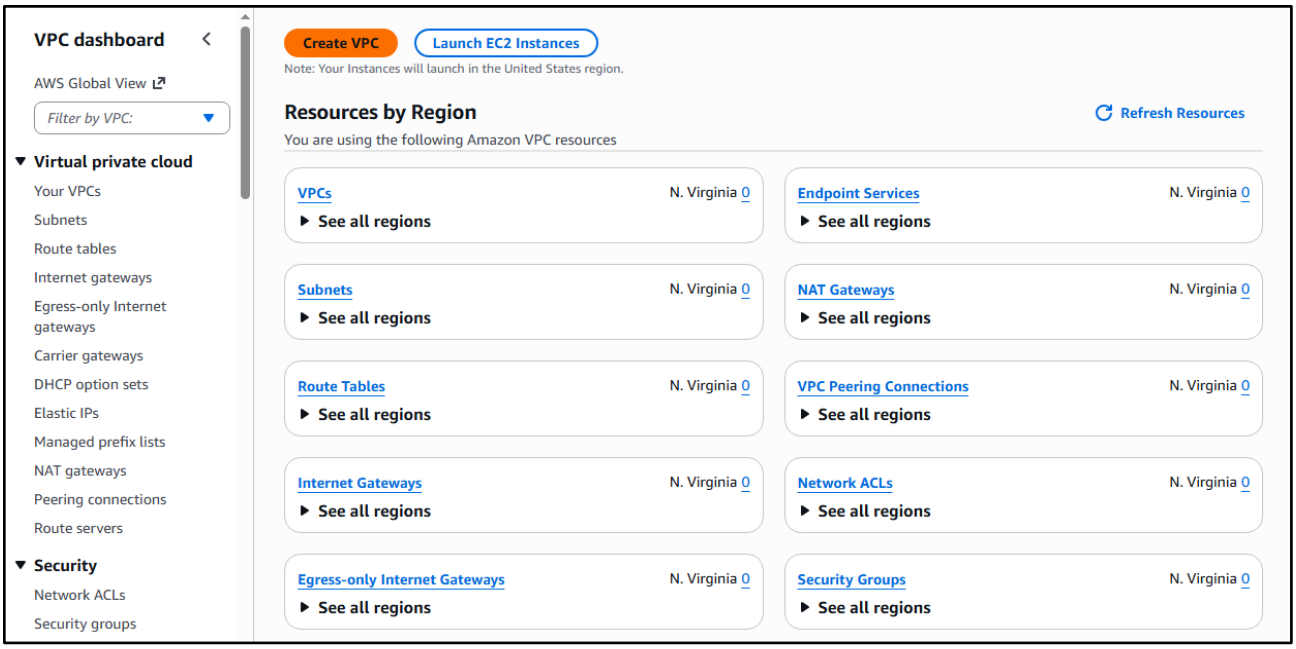

Figure 1 – VPC Dashboard

The AWS VPC dashboard is opened, showing available networking resources such as VPCs, subnets, route tables, internet gateways, and security groups. This is the starting point for

creating a secure virtual network.

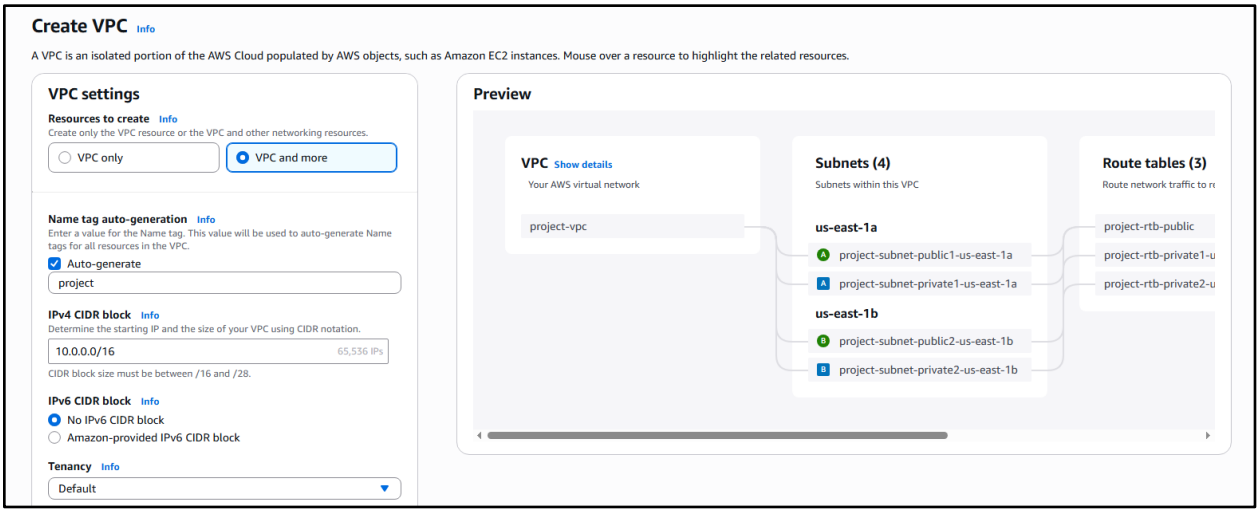

Figure 2 – Create VPC Configuration

A new VPC is created by selecting VPC and more and providing the CIDR block. The preview panel shows automatically generated subnets and route tables, confirming the VPC configuration before creation.

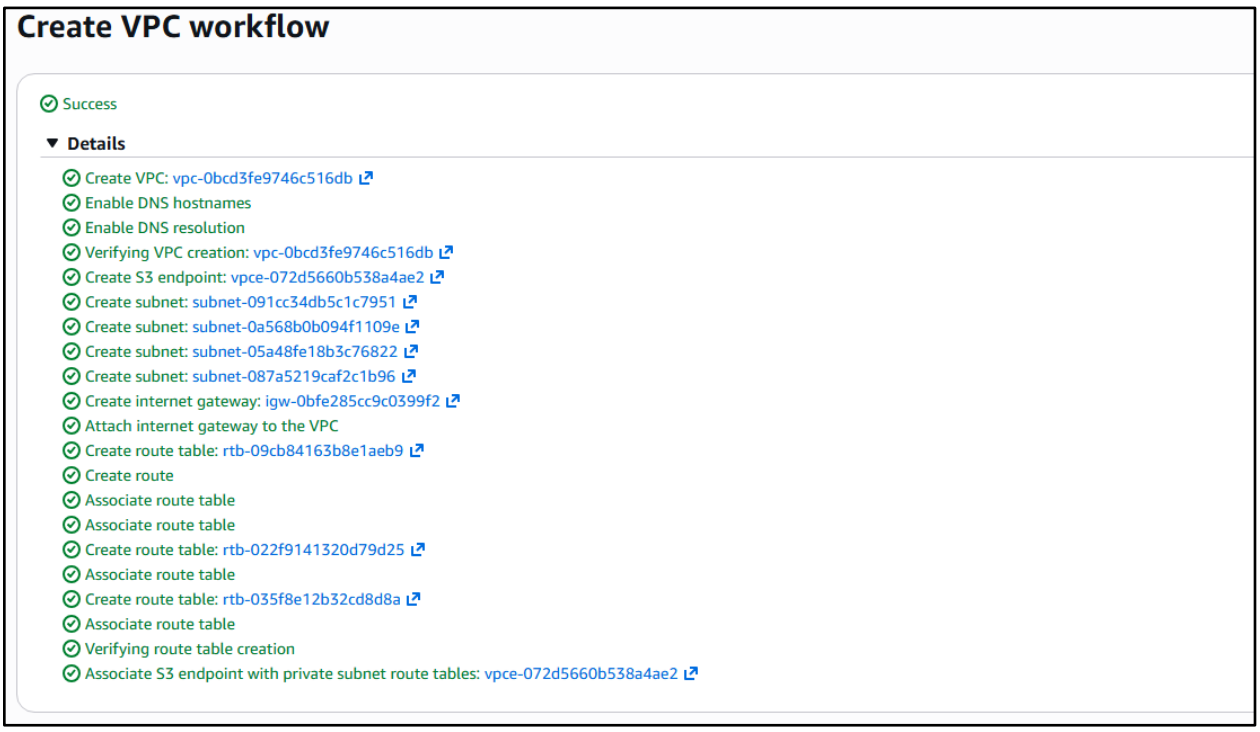

Figure 3 – VPC Creation Workflow Success

The VPC creation workflow shows a success status where AWS automatically creates subnets, route tables, internet gateway, and route associations. This confirms that the virtual network infrastructure has been successfully deployed.

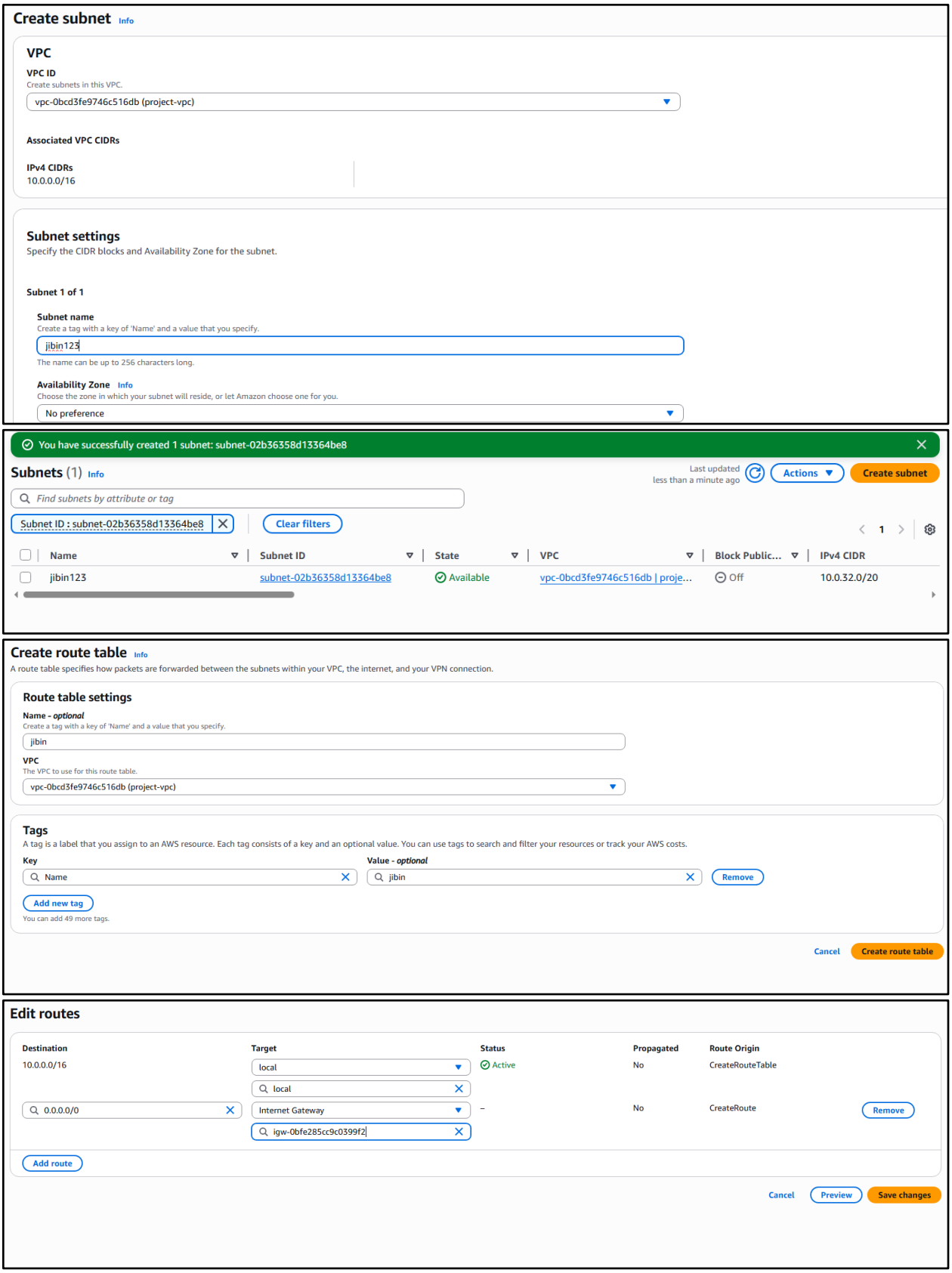

Figure 4 – Create Subnet and Route Table

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A public subnet is created inside the VPC and a route table is configured. The route table is edited to add the route 0.0.0.0/0 → Internet Gateway, enabling internet access for the subnet.

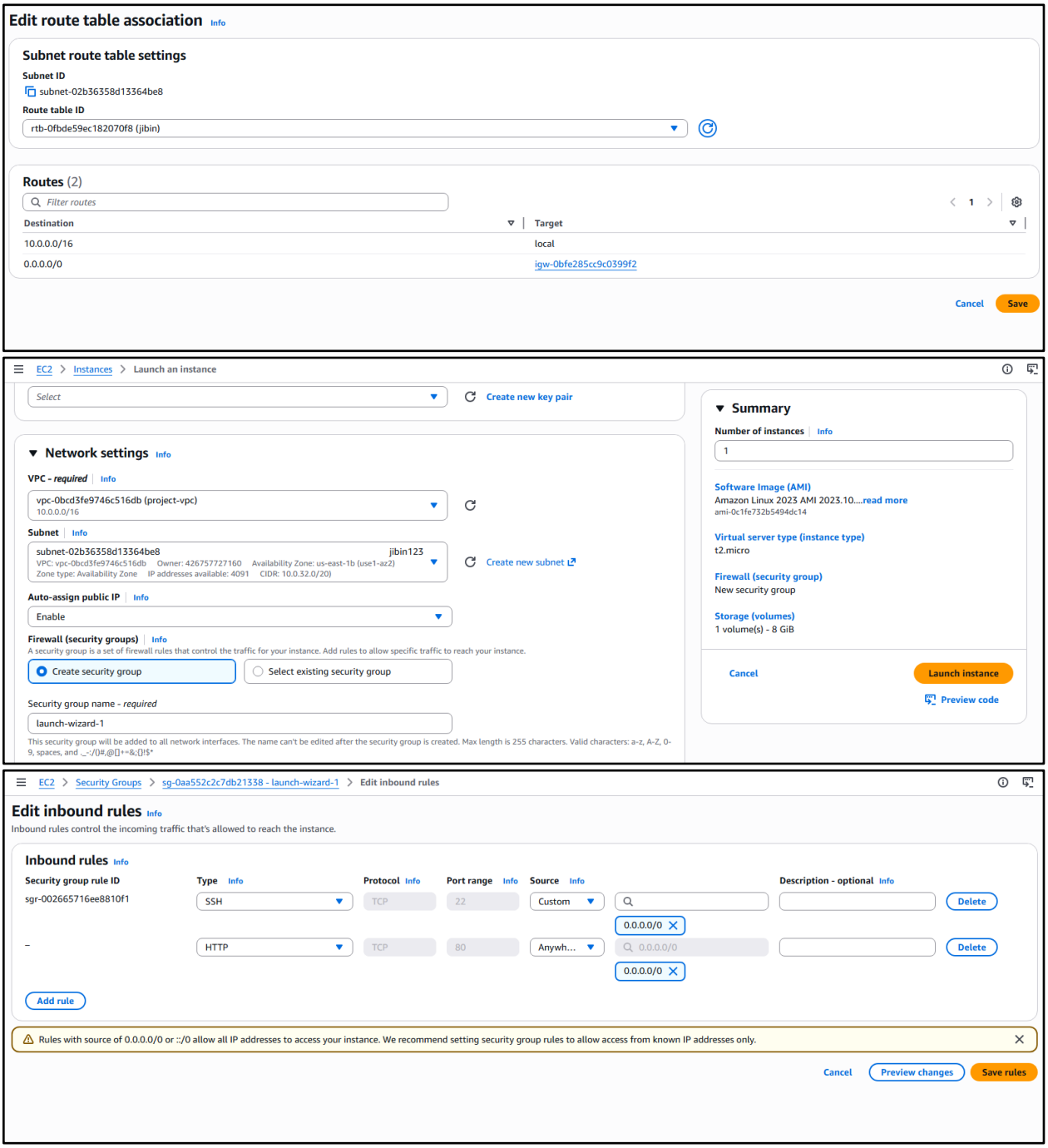

Figure 5 – Associate Route Table & Launch EC2

The subnet is associated with the route table and an EC2 instance is launched inside the created VPC and subnet. Security group inbound rules are configured to allow SSH (port 22) and HTTP (port 80) traffic.

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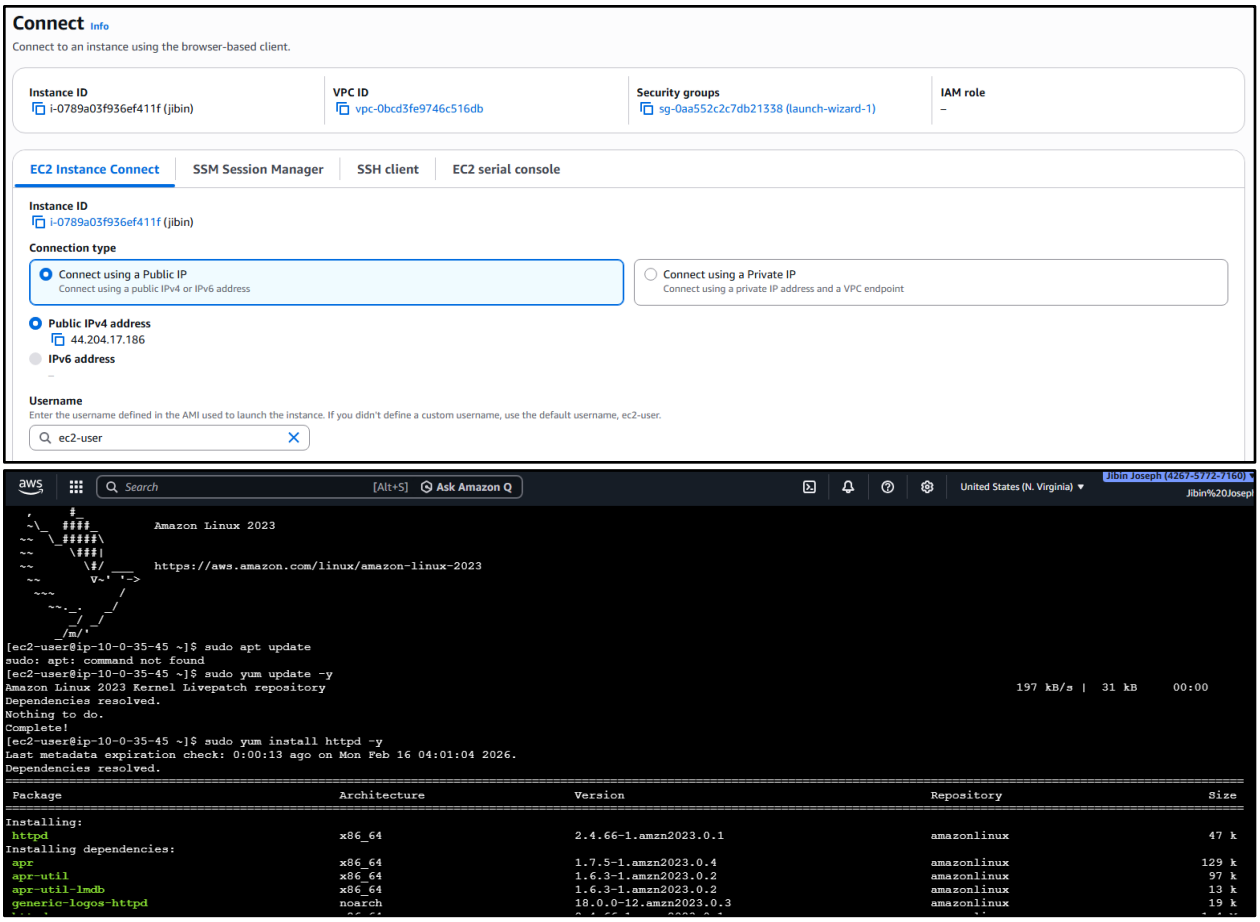

Figure 6 – Connect to EC2 and Install Web Server

The EC2 Instance Connect terminal is used to access the instance. Commands are executed to install and start the Apache web server using:

sudo yum update -y

sudo yum install httpd -y

sudo systemctl start httpd

sudo systemctl enable httpd

This prepares the server to host web pages.

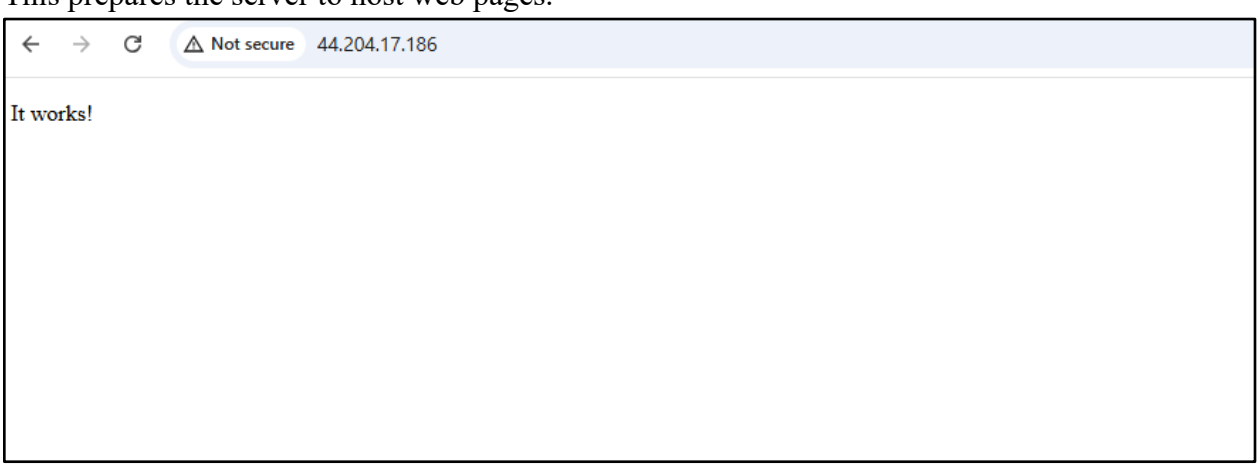

Figure 7 – Web Server Output

The public IP address of the EC2 instance is opened in a browser and the default Apache page "It works!" is displayed. This confirms successful deployment of the web server in the secure VPC environment.

2. How is SSL Certificate provided in cloud computing?

SSL certificates in AWS are provided using AWS Certificate Manager (ACM) [3].

Steps:

- Request certificate in ACM
- Validate domain ownership
- Attach certificate to Load Balancer or CloudFront
- Enable HTTPS secure communication

This encrypts communication between user and server [3]

3. What is the key pair in AWS?

A key pair is used for secure login to EC2 instances [2].

It contains:

- Public key – stored in AWS
- Private key – stored by user

Used for SSH login instead of password authentication.

Conclusion: Why cloud computing security is important.

Security as a Service (SECaaS) plays a critical role in modern cloud computing by providing built-in protection for data, applications, and infrastructure hosted in the cloud. As organizations increasingly migrate their workloads from traditional on-premise environments to cloud platforms, the need for strong security mechanisms becomes more important than ever. Through this experiment, we understood how AWS provides multiple layers of security services to protect cloud environments. Network security is ensured using Amazon VPC, subnets, route tables, internet gateways, security groups, and network ACLs, which help isolate resources and control traffic flow. Server and application security are strengthened using key pairs for secure login, firewalls for traffic filtering, and monitoring tools for detecting suspicious activities. Data security is enhanced using encryption, IAM access control, backup mechanisms, and SSL certificates that secure communication between users and servers.

References:

[1] https://cloud.google.com/security
[2] https://aws.amazon.com/security/
[3] https://docs.aws.amazon.com/security/